
Modeling and Simulation of Dynamic Systems

Project Title: Modeling and Simulation of a Four-Cylinder Engine

Mechanism Dynamics

Report

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Abstract

This project focuses on the modeling, simulation, and analysis of the dynamics of a four-cylinder internal combustion engine mechanism, specifically its core kinematic components: pistons, connecting rods, and the crankshaft. Using MATLAB and Simulink, the reciprocating motion of a piston and the resulting rotational motion of the crankshaft, driven by an electric motor model, are simulated. The report details the mathematical modeling of the DC motor driving the system and the piston kinematics. The Simulink environment facilitates the visualization and analysis of the system's behavior, particularly piston displacement over time. This model serves as an educational tool for understanding the fundamental principles of mechanical power conversion and kinematic analysis in engine systems. While a full four-cylinder combustion model is not implemented, the single-cylinder kinematics driven by a motor provides a foundational understanding applicable to more complex engine simulations.

1. Introduction

Internal Combustion Engines (ICEs) are pivotal in modern transportation and power generation, converting chemical energy from fuel into mechanical work. A key component of ICEs is the slider-crank mechanism, which transforms the linear, reciprocating motion of pistons into the rotational motion of the crankshaft. Understanding the dynamics of this mechanism is crucial for engine design, performance analysis, and efficiency optimization.

1.1. Project Objectives

The primary objectives of this project are:

- To develop a mathematical model representing the dynamics of a DC motor-driven slider-crank mechanism, analogous to one cylinder of an engine.
- To implement this model in MATLAB and Simulink for dynamic simulation.
- To analyze the simulated piston kinematics, focusing on displacement.
- To provide an educational representation of engine dynamics for studying mechanical power conversion and kinematic analysis.
- To discuss the relationship between the single-cylinder model and a complete four-cylinder engine.

1.2. Scope and Limitations

This project models the kinematics of a single piston-crank mechanism driven by a DC motor. The 3D CAD model (Figure 1) illustrates a four-cylinder assembly, but the dynamic simulation focuses on one representative cylinder.

Limitations include:

- The model is driven by an electric motor, not by combustion forces. Thus, torque generation within the cylinder is not modeled.

- A simplified constant load torque is assumed for the motor.
- Thermal effects, fluid dynamics of combustion, and detailed friction models beyond viscous damping and a constant load are not included.
- The analysis primarily focuses on piston displacement.

2. System Description

The system under consideration is based on a typical four-cylinder inline engine configuration.

2.1. Components of a Four-Cylinder Engine

A four-cylinder engine, as depicted in the 3D CAD model (Figure 1), primarily consists of:

- **Pistons:** Cylindrical components that reciprocate within the cylinders.
- **Connecting Rods:** Link the pistons to the crankshaft.
- **Crankshaft:** Converts the linear motion of the pistons into rotational motion. It has "throws" or "crankpins" offset from the main axis.
- **Cylinders/Engine Block:** Houses the pistons and forms the combustion chambers.

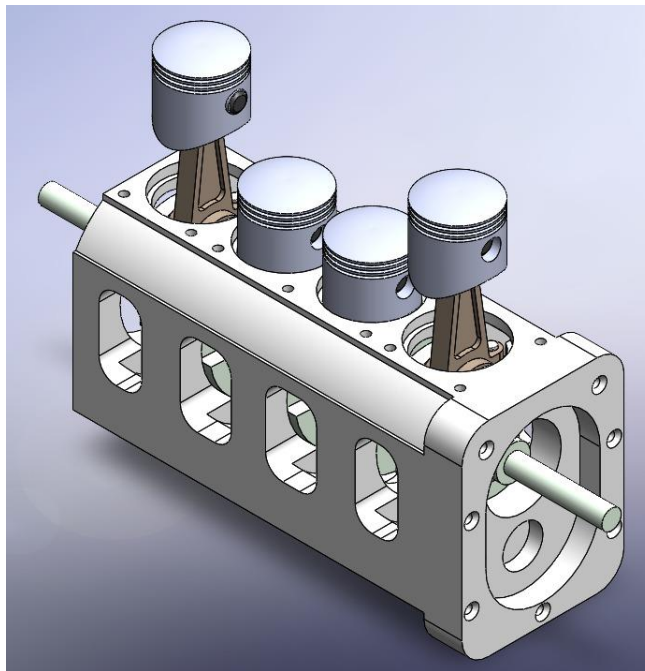


Figure 1: 3D CAD Model of a Four-Cylinder Engine Assembly

3. Mathematical Modeling

The system is modeled as a DC motor driving a single slider-crank mechanism.

3.1. DC Motor Dynamics

The DC motor is described by its electrical and mechanical equations:

- **Electrical Equation:**

$$V_a = I_a * R_a + L_a * (dI_a/dt) + E_b \quad (1)$$

Where:

- V_a is the armature voltage.
- I_a is the armature current.
- R_a is the armature resistance.
- L_a is the armature inductance.
- E_b is the back electromotive force (back-EMF).

- **Back-EMF Equation:**

$$E_b = K_e * \omega_m \quad (2)$$

Where:

- K_e is the back-EMF constant.
- ω_m is the angular velocity of the motor shaft (crankshaft).

- **Mechanical Equation:**

$$T_m = J * (d\omega_m/dt) + B * \omega_m + T_{L_crank} \quad (3)$$

Where:

- T_m is the motor torque.
- J is the total moment of inertia (motor rotor + crankshaft assembly).
- B is the viscous damping coefficient.
- T_{L_crank} is the load torque exerted by the slider-crank mechanism on the motor.
In the Simulink model, a simplified constant T_L is used.

- **Motor Torque Equation:**

$$T_m = K_t * I_a \quad (4)$$

Where:

- K_t is the motor torque constant.

3.2. Piston Kinematics

The position of the piston (x) is a function of the crank angle (θ), crank radius (r), and connecting rod length (L). The crank angle θ is the integral of the crankshaft angular velocity ω_m .

$$\theta(t) = \int \omega_m(t) dt + \theta_0 \quad (5)$$

Let ϕ be the angle the connecting rod makes with the line of piston travel. From geometric constraints:

$$L * \sin(\phi) = r * \sin(\theta) \quad (6)$$

$$\text{So, } \sin(\phi) = (r/L) * \sin(\theta) \quad (7)$$

The piston displacement x from the crankshaft axis is given by:

$$x(\theta) = r * \cos(\theta) + L * \cos(\phi) \quad (8)$$

Substituting $\cos(\phi) = \sqrt{1 - \sin^2(\phi)}$:

$$x(\theta) = r * \cos(\theta) + L * \sqrt{1 - (r/L)^2 * \sin^2(\theta)} \quad (9)$$

The piston velocity (v) is the time derivative of x :

$$v = dx/dt = (dx/d\theta) * (d\theta/dt) = (dx/d\theta) * \omega_m \quad (10)$$

$$dx/d\theta = -r * \sin(\theta) - (r^2 / L) * (\sin(\theta)\cos(\theta)) / \sqrt{1 - (r/L)^2\sin^2(\theta)} \quad (11)$$

$$\text{So, } v(\theta, \omega_m) = -\omega_m * r * [\sin(\theta) + (r / (2L)) * \sin(2\theta) / \sqrt{1 - (r/L)^2\sin^2(\theta)}] \quad (12)$$

Piston acceleration (a) is the time derivative of v . The full expression is complex and involves $d\omega_m/dt$. For constant ω_m , it is $(d^2x/d\theta^2) * \omega_m^2$.

4. Simulation in MATLAB/Simulink

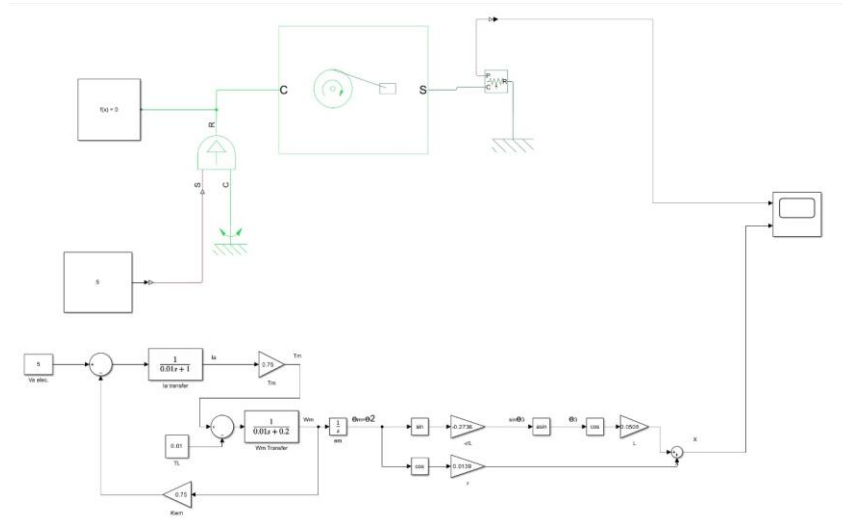


Figure 2: Simulink Model of the Motor-Driven Slider-Crank Mechanism

4.1. Model Implementation

The Simulink model consists of:

- **DC Motor Subsystem:**
 - An input voltage $V_a \text{ elec}$ (5V).
 - A summing junction calculates $V_a - E_b$.
 - A transfer function $1/(L_a*s + R_a)$ (I_a transfer) models the armature circuit to compute I_a .
 - I_a is multiplied by K_t (0.75) to get motor torque T_m .
 - Another summing junction computes $T_m - T_L$ (load torque, 0.01 N.m).
 - A transfer function $1/(J*s + B)$ (W_m Transfer) models the mechanical dynamics to compute angular velocity W_m .
 - W_m is multiplied by K_e (0.75) to compute back-EMF E_b , which is fed back.
- **Kinematic Subsystem:**
 - W_m is integrated to get crank angle θ_m .
 - Trigonometric blocks ($\sin$, $\cos$, asin) and gain blocks implement the piston displacement equation $x = r*\cos(\theta_m) + L*\cos(\phi)$, where $\phi = \text{asin}((r/L)*\sin(\theta_m))$.
 - The output X represents the piston displacement.

4.2. Simulation Parameters

The following parameters were used for the simulation, based on the Simulink model:

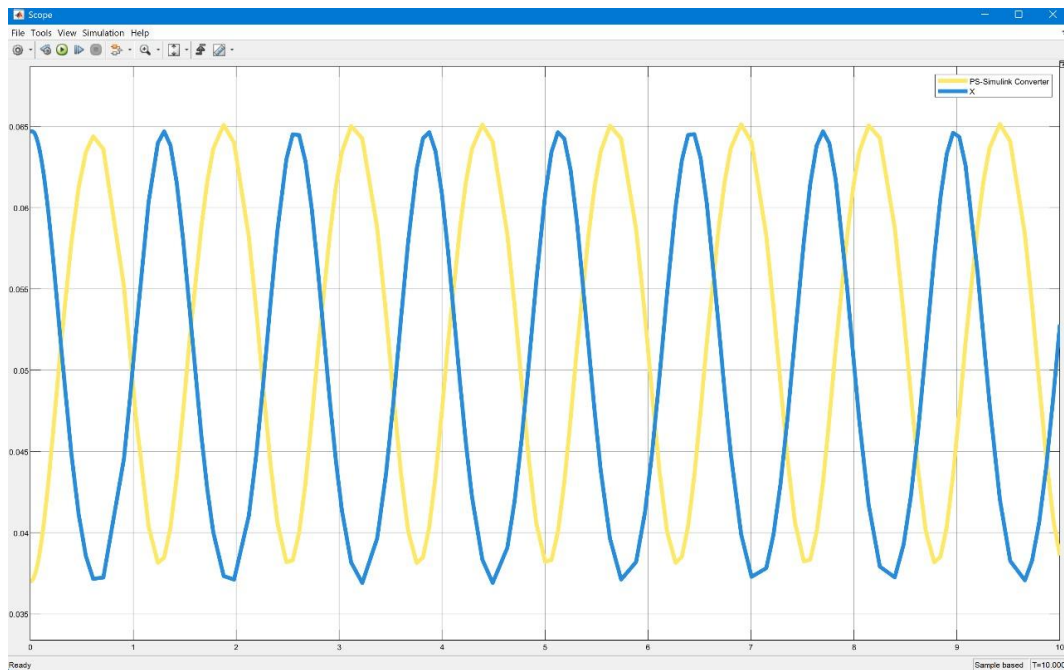
- Armature Voltage (V_a): 5 V
- Armature Resistance (R_a): 1 Ω
- Armature Inductance (L_a): 0.01 H
- Motor Torque Constant (K_t): 0.75 N.m/A
- Back-EMF Constant (K_e): 0.75 V.s/rad
- Rotor Inertia (J): 0.01 kg.m²
- Viscous Damping (B): 0.2 N.m.s/rad
- Load Torque (T_L): 0.01 N.m
- Crank Radius (r): 0.0139 m
- Connecting Rod Length (L): 0.0508 m
- Simulation Stop Time: 10.0 seconds.
- Solver: VariableStepAuto.

5. Results and Discussion

The simulation was run to observe the dynamic behavior of the system, primarily focusing on the piston displacement.

5.1. Piston Motion Analysis

Figure 3 shows the simulated piston displacement (X) over time.



5.2. Interpretation of System Behavior

The simulation successfully demonstrates the conversion of electrical energy in the motor into rotational motion of the crankshaft, which is then converted into the linear reciprocating motion of the piston. The model highlights:

- The direct relationship between crank rotation and piston linear motion.
- The non-sinusoidal nature of piston displacement due to the connecting rod's finite length.
- The dynamic response of the DC motor to an applied voltage and mechanical load.

This single-cylinder kinematic model serves as a fundamental building block. In a four-cylinder engine, four such mechanisms would be connected to the same crankshaft, typically with their crank throws phased to ensure smoother operation and power delivery. The 3D CAD model (Figure 1) helps visualize this multi-cylinder arrangement where the simulated motion would be replicated by each piston, albeit with phase differences.

6. Conclusion

This project successfully modeled and simulated a DC motor-driven slider-crank mechanism, representing the core kinematics of a single engine cylinder, using MATLAB and Simulink. The mathematical models for the DC motor and piston kinematics were implemented, and simulation results, particularly piston displacement over time, aligned with theoretical expectations. This work fulfilled its objective as an educational tool for understanding fundamental engine kinematics and electromechanical power conversion.

While the single-cylinder motor-driven model provides foundational insights, future enhancements could include multi-cylinder extensions, incorporation of combustion-based torque, and more detailed dynamic load modeling. The developed 3D CAD model visualizes the broader four-cylinder context, and the simulation provides a dynamic understanding of its individual component motion. This study lays the groundwork for more comprehensive analyses of internal combustion engine dynamics.

7. References

- Heywood, J. B. (1988). *Internal Combustion Engine Fundamentals*. McGraw-Hill. (Example reference for general ICE theory)
- Ogata, K. (2010). *Modern Control Engineering*. Prentice Hall. (Example reference for DC motor modeling)